Learning Outcome:

After completing this exercise, you shall be able to:
• Load, explore, and preprocess a categorical dataset by converting categorical features into a suitable numerical format for Deep Learning.
• Design and implement CNN and RNN models using TensorFlow/Keras for classification using the prepared dataset.
• Train Deep Learning models using different batch sizes, epochs, and optimizers and observe their training performance.
• Generate predictions and evaluate classification models using accuracy, confusion matrices, classification reports, and other relevant performance measures.
• Compare the performance of CNN and RNN models using their evaluation results and visualizations to understand their performance on the given classification dataset.
• Apply text tokenization using the Hugging Face Transformers library to prepare text data for Natural Language Processing and Generative AI applications.

Problem Statement:

In this exercise, you are required to build, train, evaluate, and compare CNN and RNN models for classification using the Mushroom/Fungi Classification (mashroom.csv) dataset. The dataset contains categorical information about different characteristics of mushrooms, such as cap shape, cap surface, odor, gill characteristics, stalk characteristics, habitat, and other features, along with a target class indicating the mushroom category.

You will load and explore the dataset, preprocess the categorical features by converting them into a suitable numerical format, and prepare the data for Deep Learning. You will then design and implement CNN and RNN models using TensorFlow/Keras, train them using the specified batch sizes, epochs, and optimizers, and generate predictions on the test data. You will evaluate and compare both models using accuracy, confusion matrices, classification reports, and suitable visualizations.
In the second part of the exercise, you will work with Generative AI and Natural Language Processing (NLP) by using the provided exercise_text.json file and the Hugging Face Transformers tokenizer to tokenize text data.

For the Deep Learning tasks, you must use TensorFlow/Keras. PyTorch is not allowed. For the text tokenization task, use the Hugging Face Transformers library.

You can use any programming environment such as Jupyter Notebook, Google Colab, VS Code, or any other Python editor to write and execute your code.

You can use the following datasets in the repo named mashroom.csv.